Project Guidelines: NLP-Driven Reinforcement Learning Trading Platform

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Overview

Build a small end-to-end intelligent trading platform that integrates **Natural Language Processing (NLP)** and **Reinforcement Learning (RL)**. The platform must demonstrate the full pipeline: *raw text → NLP signal → RL trading decision*, reflecting real-world quantitative finance practice.

Platform Modules (Each Sub-team Owns One)

Module	Responsibility	Key Tasks
① **Data Ingestion**	Collect & stream data	Fetch financial news (e.g., RSS/NewsAPI) + market OHLCV prices (e.g., yfinance); schedule automated collection; log data flow
② **NLP Pipeline**	Text → Signal	Preprocess text (tokenize, remove stopwords, lemmatize); perform sentiment analysis (lexicon-based, Logistic Regression, and FinBERT); aggregate daily sentiment scores per ticker
③ **Data Storage**	Persist everything	Design DB schema (news, market data, sentiment signals, trading logs); implement data access layer using SQLite / PostgreSQL
④ **RL Trading Engine**	Learn to trade	Build FinancialTradingEnv (state: price indicators + sentiment score; actions: Buy/Sell/Hold); implement DQN from scratch with experience replay and target network; train and evaluate
⑤ **Front-End**	Visualize & monitor	Display live sentiment trends, agent decisions, portfolio performance, and system health using

Module	Responsibility	Key Tasks
Dashboard		Streamlit or Flask

Core Integration Requirement

The **sentiment score** from Module ② must be included in the **RL state vector** in Module ④. Students must run an **ablation study** (with vs. without NLP signal) to answer: *"Do NLP signals improve RL trading performance?"*

Deliverables

1. **Working Platform** — All 5 modules integrated and runnable from a single entry point
2. **Written Report** (25–50 pages) — Cover background, methodology, system design, experiments, results, and critical reflection for each module
3. **GitHub Repository** — Clean code with README.md, requirements.txt, and module-level documentation
4. **Group Presentation** (25 min + Q&A) — Live demo of the platform required

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Evaluation Criteria

Category	Weight	Focus
NLP Pipeline Quality	25%	Preprocessing correctness; comparison of ≥2 sentiment methods; F_1 score
RL Agent Design & Training	25%	DQN implementation; reward function justification; convergence analysis
Platform Integration	20%	Data flows correctly across all 5 modules; real-time or batch pipeline works end-to-end

Category	Weight	Focus
Report & Analysis	20%	Ablation study; performance metrics (Sharpe Ratio, MDD); critical discussion
Presentation & Demo	10%	Clarity; live demo; ability to answer technical questions

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Minimum Technical Requirements

- **NLP:** Implement at least **two** sentiment approaches; use FinBERT as one of them
- **RL:** Implement DQN **from scratch** (no Stable-Baselines3); minimum 200 training episodes
- **State vector:** Must include at least: [price, MA50, MA200, RSI, MACD, position, cash, sentiment_score]
- **Evaluation:** Report Sharpe Ratio, Max Drawdown (MDD), and compare against a buy-and-hold benchmark
- **Validation:** Apply **walk-forward validation** — no look-ahead bias permitted

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Group size: 1–6 students | Duration: 8 weeks | Deadline: 14-th weekend